

Representing Real-Life Linear Relationships

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A **linear relationship** shows a constant rate of change between two variables.

Representations

Linear relationships can be represented using a **table of values**, a **graph**, an **equation**, and a **verbal description**.

Linear Equation

The general form of a linear equation is

$$y = mx + b$$

where m is the **slope** and b is the **y-intercept**.

Slope

The **slope** represents the rate of change in a real-life situation.

Examples

Example 1

A taxi charges a base fare of ₱50 plus ₱15 per kilometer.

Write a linear equation for the situation.

Example 1

The base fare is the initial value:

$$b = 50$$

The charge per kilometer is the rate of change:

$$m = 15$$

Therefore,

$$\boxed{y = 15x + 50}$$

Example 2

A cellphone plan costs ₱300 plus ₱2 per text message.

Write a linear equation for the situation.

Example 2

The initial cost is

$$b = 300$$

The cost per text message is

$$m = 2$$

Therefore,

$$\boxed{y = 2x + 300}$$

Example 3

The temperature increases by 2°C every hour starting from 18°C .

Write a linear equation for the situation.

Example 3

The starting temperature is

$$b = 18$$

The rate of change is

$$m = 2$$

Therefore,

$$\boxed{y = 2x + 18}$$

Example 4

A water tank contains 500 L and drains at 40 L per minute.

Write a linear equation for the situation.

Example 4

The initial amount is

$$b = 500$$

Since the tank is draining, the rate is negative:

$$m = -40$$

Therefore,

$$\boxed{y = -40x + 500}$$

Example 5

A straight-line graph passes through $(0, 4)$ and $(2, 10)$.

Write its linear equation.

Example 5

Find the slope:

$$m = \frac{10 - 4}{2 - 0} = \frac{6}{2} = 3$$

Since the graph passes through $(0, 4)$,

$$b = 4$$

Therefore,

$$\boxed{y = 3x + 4}$$

Practice Exercises

Practice 1

A taxi charges a base fare of ₱50 plus ₱15 per kilometer.

Write a linear equation based on the situation.

Practice 2

A cellphone plan costs ₱300 plus ₱2 per text message.

Write a linear equation based on the situation.

Practice 3

A student earns ₱120 per hour tutoring.

Write a linear equation based on the situation.

Practice 4

The temperature increases by 2°C every hour starting from 18°C .

Write a linear equation based on the situation.

Practice 5

A car travels at a constant speed of 60 km/h.
Write a linear equation based on the situation.

Practice 6

A movie rental store charges ₱100 plus ₱25 per day.

Write a linear equation based on the situation.

Practice 7

A water tank contains 500 L and drains at 40 L per minute.

Write a linear equation based on the situation.

Practice 8

The cost C of apples is ₱80 per kilogram.
Write a linear equation based on the situation.

Practice 9

A gym membership costs ₱1500 plus ₱300 per month.

Write a linear equation based on the situation.

Practice 10

A straight-line graph passes through $(0, 4)$ and $(2, 10)$.

Write a linear equation based on the situation.

Activities

Activity 1

A delivery service charges a base fee of ₱40 plus ₱18 per kilometer.

Write a linear equation.

Activity 2

A streaming subscription costs ₱250 plus ₱5 per downloaded movie.

Write a linear equation.

Activity 3

A worker earns ₱95 per hour.
Write a linear equation.

Activity 4

The temperature decreases by 3°C every hour starting from 30°C .

Write a linear equation.

Activity 5

A train travels at a constant speed of 80 km/h.
Write a linear equation.

Activity 6

A bike rental shop charges ₱200 plus ₱35 per hour.

Write a linear equation.

Activity 7

A tank contains 900 L of water and fills at 50 L per minute.

Write a linear equation.

Activity 8

The cost C of rice is ₱55 per kilogram.
Write a linear equation.

Activity 9

A registration fee is ₱2000 plus ₱450 per event attended.

Write a linear equation.

Activity 10

A straight-line graph passes through $(0, -2)$ and $(4, 14)$.

Write a linear equation.

Thank you!